

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Switcher
Manufacturer: Extron
Firmware Version: 2.01
Model(s): MPS 602, MPS 602 MA, MPS 602 SA

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0001-b006	1.4.2

Version History

Module Version	Date	Notes
1_2_0_0	3/14/2018	Renamed Input Signal to Input Signal Status and states to Active and Not Active. Renamed HDCP Authorization to HDCP Input Authorization. Renamed Audio Gain/Attenuation to Input Gain. Removed EDID (DVI), EDID (HDMI) and EDID (VGA). Renamed Mic Talk-Over Threshold to Mic Talk Over Threshold. Renamed Auto Switch Mode States Highest/Lowest Priority to Highest/Lowest Active Input.
1_1_2_0	10/12/2017	Updated module to Rev. B.
1_1_1_2	7/15/2016	Update to current standards.

Global Scripter Module Communication Sheet

1_1_1_1	4/16/2016	Update Connection Status.
1_1_1_0	2/11/2016	Initial Version.

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='MPS 602')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='MPS 602')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
AudioInputFormat	'Embedded'	'Analog'	'Auto'
Qualifier Key	Qualifier Value		
'Input'	'3' – '6'		
# AudioInputFormat example InterfaceName.Set('AudioInputFormat', 'Embedded', {'Input': '3'})			
Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command	Value	Value	Value
AutoSwitchMode	'Highest Active Input'	'Lowest Active Input'	'Off'
# AutoSwitchMode example InterfaceName.Set('AutoSwitchMode', 'Highest Active Input')			
Command	Value	Value	Value
ExecutiveMode	'Mode 1'	'Mode 2'	'Off'
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'Mode 1')			
Command	Value	Value	
HDCPInputAuthorization	'On'	'Off'	
Qualifier Key	Qualifier Value		
'Input'	'3' – '6'		
# HDCPInputAuthorization example InterfaceName.Set('HDCPInputAuthorization', 'On', {'Input': '3'})			
Command	Value		
Input	'1' – '6'		
Qualifier Key	Qualifier Value	Qualifier Value	Qualifier Value
'Tie Type'	'Audio/Video'	'Video'	'Audio'
# Input example InterfaceName.Set('Input', '1', {'Tie Type': 'Audio/Video'})			
Command	Value		
InputGain	-18 to 24 in steps of 1		
Qualifier Key	Qualifier Value		
'Input'	'1' – '6'		

# InputGain example InterfaceName.Set('InputGain', 24, {'Input': '1'})		
Command MicGain	Value -18 to 60 in steps of 1	
# MicGain example InterfaceName.Set('MicGain', 60)		
Command MicMute	Value 'On'	Value 'Off'
# MicMute example InterfaceName.Set('MicMute', 'On')		
Command MicTalkOverThreshold	Value 0 to 30 in steps of 1	
# MicTalkOverThreshold example InterfaceName.Set('MicTalkOverThreshold', 30)		
Command ProgramAudioDucking	Value 0 to 30 in steps of 1	
# ProgramAudioDucking example InterfaceName.Set('ProgramAudioDucking', 30)		
Command RGBBreakaway	Value '1' – '2'	
# RGBBreakaway example InterfaceName.Set('RGBBreakaway', '1')		
Command VideoMute	Value 'On'	Value 'Off'
# VideoMute example InterfaceName.Set('VideoMute', 'On')		
Command Volume	Value 0 to 100 in steps of 1	
# Volume example InterfaceName.Set('Volume', 100)		

Status Available

For all commands, Update should be called only once since the command's status will be updated automatically as the device's status changes. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AudioInputFormat	'Embedded'	'Analog'	'Auto'
Qualifier Key	Qualifier Value		
'Input'	'3' – '6'		
# AudioInputFormat examples InterfaceName.Update('AudioInputFormat', {'Input': '3'}) Value = InterfaceName.ReadStatus('AudioInputFormat', {'Input': '3'}) InterfaceName.SubscribeStatus('AudioInputFormat', None, FeedbackHandler)			
Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	Value
AutoSwitchMode	'Highest Active Input'	'Lowest Active Input'	'Off'
# AutoSwitchMode examples InterfaceName.Update('AutoSwitchMode') Value = InterfaceName.ReadStatus('AutoSwitchMode') InterfaceName.SubscribeStatus('AutoSwitchMode', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
ExecutiveMode	'Mode 1'	'Mode 2'	'Off'
# ExecutiveMode examples			

<pre>InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)</pre>			
Command HDCPInputAuthorization ¹	Value 'On'	Value 'Off'	
Qualifier Key 'Input'	Qualifier Value '3' – '6'		
<pre># HDCPInputAuthorization examples InterfaceName.Update('HDCPInputAuthorization') Value = InterfaceName.ReadStatus('HDCPInputAuthorization', {'Input': '3'}) InterfaceName.SubscribeStatus('HDCPInputAuthorization', None, FeedbackHandler)</pre>			
Command HDCPInputStatus ¹	Value 'No Source Connected'	Value 'Source Connected and HDCP'	Value 'Source Connected and No HDCP'
Qualifier Key 'Input'	Qualifier Value '3' – '6'		
<pre># HDCPInputStatus examples InterfaceName.Update('HDCPInputStatus') Value = InterfaceName.ReadStatus('HDCPInputStatus', {'Input': '3'}) InterfaceName.SubscribeStatus('HDCPInputStatus', None, FeedbackHandler)</pre>			
Command HDCPOutputStatus	Value 'No Sync Detected'	Value 'Sync Detected with HDCP'	Value 'Sync Detected with No HDCP'
<pre># HDCPOutputStatus examples InterfaceName.Update('HDCPOutputStatus') Value = InterfaceName.ReadStatus('HDCPOutputStatus') InterfaceName.SubscribeStatus('HDCPOutputStatus', None, FeedbackHandler)</pre>			
Command Input ¹	Value '1' – '6'		
Qualifier Key 'Tie Type'	Qualifier Value 'Audio/Video'	Qualifier Value 'Video'	Qualifier Value 'Audio'
<pre># Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input', {'Tie Type': 'Audio/Video'}) InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command InputGain	Value -18 to 24 in steps of 1		
Qualifier Key 'Input'	Qualifier Value '1' – '6'		
<pre># InputGain examples InterfaceName.Update('InputGain', {'Input': '1'}) Value = InterfaceName.ReadStatus('InputGain', {'Input': '1'}) InterfaceName.SubscribeStatus('InputGain', None, FeedbackHandler)</pre>			
Command	Value	Value	

Global Scripter Module
Communication Sheet

InputSignalStatus ¹	'Active'	'Not Active'
Qualifier Key 'Input'	Qualifier Value '1' – '6'	
# InputSignalStatus examples InterfaceName.Update('InputSignalStatus') Value = InterfaceName.ReadStatus('InputSignalStatus', {'Input': '1'}) InterfaceName.SubscribeStatus('InputSignalStatus', None, FeedbackHandler)		
Command MicGain	Value -18 to 60 in steps of 1	
# MicGain examples InterfaceName.Update('MicGain') Value = InterfaceName.ReadStatus('MicGain') InterfaceName.SubscribeStatus('MicGain', None, FeedbackHandler)		
Command MicMute	Value 'On'	Value 'Off'
# MicMute examples InterfaceName.Update('MicMute') Value = InterfaceName.ReadStatus('MicMute') InterfaceName.SubscribeStatus('MicMute', None, FeedbackHandler)		
Command MicTalkOverThreshold	Value 0 to 30 in steps of 1	
# MicTalkOverThreshold examples InterfaceName.Update('MicTalkOverThreshold') Value = InterfaceName.ReadStatus('MicTalkOverThreshold') InterfaceName.SubscribeStatus('MicTalkOverThreshold', None, FeedbackHandler)		
Command ProgramAudioDucking	Value 0 to 30 in steps of 1	
# ProgramAudioDucking examples InterfaceName.Update('ProgramAudioDucking') Value = InterfaceName.ReadStatus('ProgramAudioDucking') InterfaceName.SubscribeStatus('ProgramAudioDucking', None, FeedbackHandler)		
Command RGBBreakaway	Value '1' – '2'	
# RGBBreakaway examples InterfaceName.Update('RGBBreakaway') Value = InterfaceName.ReadStatus('RGBBreakaway') InterfaceName.SubscribeStatus('RGBBreakaway', None, FeedbackHandler)		
Command VideoMute	Value 'On'	Value 'Off'
# VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)		
Command Volume	Value 0 to 100 in steps of 1	


```
# Volume examples  
InterfaceName.Update('Volume')  
Value = InterfaceName.ReadStatus('Volume')  
InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)
```

¹ Qualifier parameters is not required when calling the Update method with this command because the response returns statuses for all qualifier states

Cable and Adapter Requirements

Captive Screw to Captive Screw

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

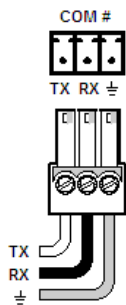
Data Bits: 8

Parity: None

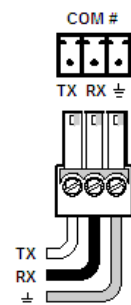
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Signal
TxD	→	RxD
RxD	←	TxD
GND		GND



Appendix A. Set Commands

Audio Mute Off	0Z
Audio Mute On	1Z
Auto Switch Mode Highest Active Input	W1AUSW\x0D
Auto Switch Mode Lowest Active Input	W2AUSW\x0D
Auto Switch Mode Off	W0AUSW\x0D
Executive Mode Mode 1	1X
Executive Mode Mode 2	2X
Executive Mode Off	0X
Input 0 Tie Type Audio	0\$
Input 6 Tie Type Audio	6\$
Input 0 Tie Type Audio/Video	0!
Input 6 Tie Type Audio/Video	6!
Input 0 Tie Type Video	0%
Input 6 Tie Type Video	6%
Input Gain -18 Input 1	1*18g
Input Gain 24 Input 1	1*24G
Input Gain -18 Input 6	6*18g
Input Gain 24 Input 6	6*24G
Mic Mute Off	0M
Mic Mute On	1M
Mic Talk Over Threshold 0	0*2#
Mic Talk Over Threshold 30	30*2#
Program Audio Ducking 0	0*58#
Program Audio Ducking 30	30*58#
RGB Breakaway 1	1&
RGB Breakaway 2	2&
Video Mute Off	0B
Video Mute On	1B
Volume 0	0V
Volume 100	100V

Appendix B. Update Commands

Audio Mute	Z
Auto Switch Mode	WAUSW\x0D
Executive Mode	X
HDCP Input Authorization Input 3	WEHDCP\x0D
HDCP Input Authorization Input 6	WEHDCP\x0D
HDCP Input Status Input 3	WIHDCP\x0D
HDCP Input Status Input 6	WIHDCP\x0D
HDCP Output Status	WOHDCP\x0D
Input Signal Status Input 1	W0LS\x0D
Input Signal Status Input 6	W0LS\x0D
Mic Mute	M
RGB Breakaway	&
Volume	V